

Practical Lab Sheet

GitHub Runner Security & Management on Windows

Aim	To install, test, secure, manage, restart, and remove a GitHub self-hosted runner on Windows.
Platform	Windows 10 / Windows 11
Tools Required	GitHub account, Internet connection, Command Prompt, GitHub repository
Main Concept	A self-hosted runner is your own machine used to run GitHub Actions jobs.

Important instruction for students: Follow the steps exactly in order. Do not close the GitHub runner page or Command Prompt until the practical asks you to do so.

Part A: Create a GitHub Repository

Step 1: Open Google Chrome or any browser.

Step 2: Go to GitHub.

Step 3: Sign in with your GitHub account.

Step 4: Click the + icon at the top-right side.

Step 5: Click New repository.

Step 6: In Repository name, type:

runner-security-lab

Step 7: Select Public.

Step 8: Click Create repository.

Part B: Open the Self-Hosted Runner Page

Step 1: Open your repository named runner-security-lab.

Step 2: Click Settings.

Step 3: Click Actions from the left-side menu.

Step 4: Click Runners.

Step 5: Click New self-hosted runner.

Step 6: Select Windows.

Step 7: Select x64.

Step 8: Now GitHub will show three command sections: Download, Configure, and Run.

Step 9: Do not close this GitHub page.

Part C: Open Command Prompt as Administrator

Step 1: Click the Windows search bar.

Step 2: Type:

cmd

Step 3: Right-click Command Prompt.

Step 4: Click Run as administrator.

Step 5: Click Yes if Windows asks for permission.

Part D: Create Runner Folder in C Drive

Step 1: Run this command to go to C drive:

```
cd C:\
```

Step 2: Run this command to create a folder:

```
mkdir actions-runner
```

Step 3: Run this command to open the folder:

```
cd actions-runner
```

Part E: Download GitHub Runner

Step 1: Go back to the GitHub runner page.

Step 2: Find the Download section.

Step 3: Copy the download command shown by GitHub.

Step 4: Paste it in Command Prompt.

Step 5: Press Enter.

Step 6: Wait until the download is complete.

Example only. Your command may have a different version number:

```
Invoke-WebRequest -Uri https://github.com/actions/runner/releases/download/v2.xxx.x/actions-runner-win-x64-2.xxx.x.zip -OutFile actions-runner-win-x64.zip
```

Part F: Extract GitHub Runner ZIP File

Step 1: Go to the GitHub runner page.

Step 2: Find the Extract section.

Step 3: Copy the extract command shown by GitHub.

Step 4: Paste it in Command Prompt.

Step 5: Press Enter.

Step 6: Wait until extraction is complete.

Example command:

```
Add-Type -AssemblyName System.IO.Compression.FileSystem ;  
[System.IO.Compression.ZipFile]::ExtractToDirectory("$PWD/actions-runner-win-x64.zip", "$PWD")
```

Part G: Configure the Runner

Step 1: Go to the GitHub runner page.

Step 2: Find the Configure section.

Step 3: Copy the configure command shown by GitHub.

Step 4: Paste it in Command Prompt.

Step 5: Press Enter.

Example only. Your URL and token will be different:

```
.\config.cmd --url https://github.com/YOUR-USERNAME/runner-security-lab --token YOUR-TOKEN
```

Answer the Configuration Questions

Question on Screen	Student Action / Answer
Enter the name of the runner group to add this runner to:	Press Enter
Enter the name of runner:	student-runner
Enter any additional labels:	self-hosted,windows,student
Enter name of work folder:	Press Enter
Run as service?	N

Part H: Start the Runner

Step 1: Run this command in Command Prompt:

```
.\run.cmd
```

Step 2: Keep Command Prompt open.

Step 3: Do not close this window during the practical.

Step 4: You should see this type of message:

```
Listening for Jobs
```

Part I: Create First GitHub Actions Workflow

Step 1: Go to your GitHub repository.

Step 2: Click Add file.

Step 3: Click Create new file.

Step 4: In file name box, type:

```
.github/workflows/runner-test.yml
```

Step 5: Paste this YAML code:

```
name: Runner Test on Windows

on:
  workflow_dispatch:

jobs:
  test:
    runs-on: self-hosted

    steps:
      - name: Print message
        run: echo Runner is working successfully

      - name: Show computer name
        run: hostname

      - name: Show current user
        run: whoami
```

```
- name: Show current directory
  run: cd
```

Step 6: Click Commit changes.

Part J: Run the Workflow

Step 1: Click Actions in your repository.

Step 2: Click Runner Test on Windows.

Step 3: Click Run workflow.

Step 4: Click the green Run workflow button.

Step 5: Wait for the job to start.

Step 6: Look at Command Prompt.

Step 7: You will see the job running on your Windows machine.

Step 8: After completion, GitHub will show a green tick.

Part K: Create Runner Security Workflow

Step 1: Go to your GitHub repository.

Step 2: Click Add file.

Step 3: Click Create new file.

Step 4: In file name box, type:

```
.github/workflows/runner-security.yml
```

Step 5: Paste this YAML code:

```
name: Runner Security Check
```

```
on:
```

```
  workflow_dispatch:
```

```
permissions:
```

```
  contents: read
```

```
jobs:
```

```
  security-check:
```

```
    runs-on: self-hosted
```

```
  steps:
```

```
    - name: Security message
```

```
      run: echo Always use minimum permissions on self-hosted runners
```

```
    - name: Check user
```

```
      run: whoami
```

```
    - name: Check machine
```

```
      run: hostname
```

```
- name: Check folder
  run: cd
```

Step 6: Click Commit changes.

Step 7: Click Actions.

Step 8: Click Runner Security Check.

Step 9: Click Run workflow.

Step 10: Click green Run workflow button.

Step 11: Check that the job completes successfully.

Part L: Understand the Security Setting

This line gives the GitHub Actions token only read access to repository contents:

```
permissions:
  contents: read
```

Do not use this in student labs unless it is required:

```
permissions: write-all
```

Reason: write-all gives too much permission. Minimum permission is safer for self-hosted runners.

Part M: Check Runner Status

Step 1: Open your GitHub repository.

Step 2: Click Settings.

Step 3: Click Actions.

Step 4: Click Runners.

Step 5: Check the runner name:

```
student-runner
```

Step 6: The status should be Idle or Active.

Part N: Stop the Runner

Step 1: Go to Command Prompt where runner is running.

Step 2: Press this key combination:

```
Ctrl + C
```

Step 3: If it asks to terminate batch job, type:

```
Y
```

Step 4: Press Enter.

Step 5: The runner is now stopped.

Part O: Start the Runner Again

Step 1: Run this command:

```
cd C:\actions-runner
```

Step 2: Run this command:

```
.\run.cmd
```

Step 3: The runner will start again.

Step 4: You should again see Listening for Jobs.

Part P: Remove the Runner

Step 1: Go to your GitHub repository.

Step 2: Click Settings.

Step 3: Click Actions.

Step 4: Click Runners.

Step 5: Click student-runner.

Step 6: Click Remove.

Step 7: Copy the remove token or remove command shown by GitHub.

Step 8: Go to Command Prompt.

Step 9: Stop the runner if it is running by pressing Ctrl + C.

Step 10: Run this command:

```
cd C:\actions-runner
```

Step 11: Run this command:

```
.\config.cmd remove
```

Step 12: Paste the removal token if GitHub asks for it.

Step 13: Press Enter.

Step 14: The runner is removed from GitHub.

Important Security Rules for Students

- Do not use a self-hosted runner for unknown public pull requests.
- Do not store passwords on the runner machine.
- Do not keep secret files inside C:\actions-runner.
- Use minimum workflow permission.
- Keep Windows updated.
- Stop or remove the runner after lab completion.
- Use separate runners for separate projects when possible.
- Do not share the runner token with anyone.

Common Errors and Simple Fixes

Problem	Fix
Runner not visible on GitHub	Check internet connection and run <code>.\run.cmd</code> again.
Workflow waiting for runner	Make sure Command Prompt shows Listening for Jobs.
Token expired	Go back to GitHub runner page and generate a new command/token.
Permission denied	Run Command Prompt as Administrator.
YAML file not running	Check file path: <code>.github/workflows/runner-test.yml</code>

Final Result

The GitHub self-hosted runner was successfully installed on Windows, connected with GitHub, tested using a workflow, secured using minimum permissions, stopped, restarted, and removed.

References

- GitHub Docs - Hosting your own runners: <https://docs.github.com/actions/hosting-your-own-runners>
- GitHub Docs - Adding self-hosted runners:
<https://docs.github.com/actions/hosting-your-own-runners/adding-self-hosted-runners>
- GitHub Docs - Secure use of GitHub Actions:
<https://docs.github.com/actions/security-for-github-actions/security-guides/security-hardening-for-github-actions>
- GitHub Docs - Managing self-hosted runners:
<https://docs.github.com/actions/hosting-your-own-runners/managing-self-hosted-runners>